Workbook Template

Project Name: AegisAura

Problem Statement:

Women often feel unsafe while travelling alone, working late nights, or in isolated areas due to lack of real-time help, delayed police response, and no automatic alert system. In many cases, victims are unable to manually call for help. AegisAura uses AI, GPS and IoT to provide instant SOS, live location tracking and automatic threat detection to ensure women safety.

Why does this problem exist?

- Live location tracking is not automated.
- No instant alert system to family/police.s
- Unsafe routes and isolated areas.
- Manual SOS is not possible during panic.
- Low police patrolling in certain areas.
- Lack of AI-based threat prediction(shouting, fall detection).
- Women don't now which route is safe .

Who are the stakeholders?

Internal users:

- App Administrators
- Support / Monitoring Team
- Police Control Room staff

External users:

- Women Users / Girls / Working Professionals
- Family Members / Guardians
- Police

Expected system users:

- Women Users
- Family Members
- Police Admin

Objectives of the project:

- To provide one-click SOS alert to family and police.
- To track live location in real-time using GPS.
- To detect threats automatically using AI – like loud scream, sudden fall.
- To suggest safest route using AI & crime data.
- To record audio/video evidence automatically.
- To reduce emergency response time.
- To create a secure community for women.

Expected Outputs:

- Instant SOS alert with live location.
- Real-time location tracking on family dashboard .
- Automatic threat detection.
- Safe route suggestion.
- Audio/Video evidence recording.
- Near by police station & help line connection.
- Better sense of security for women.

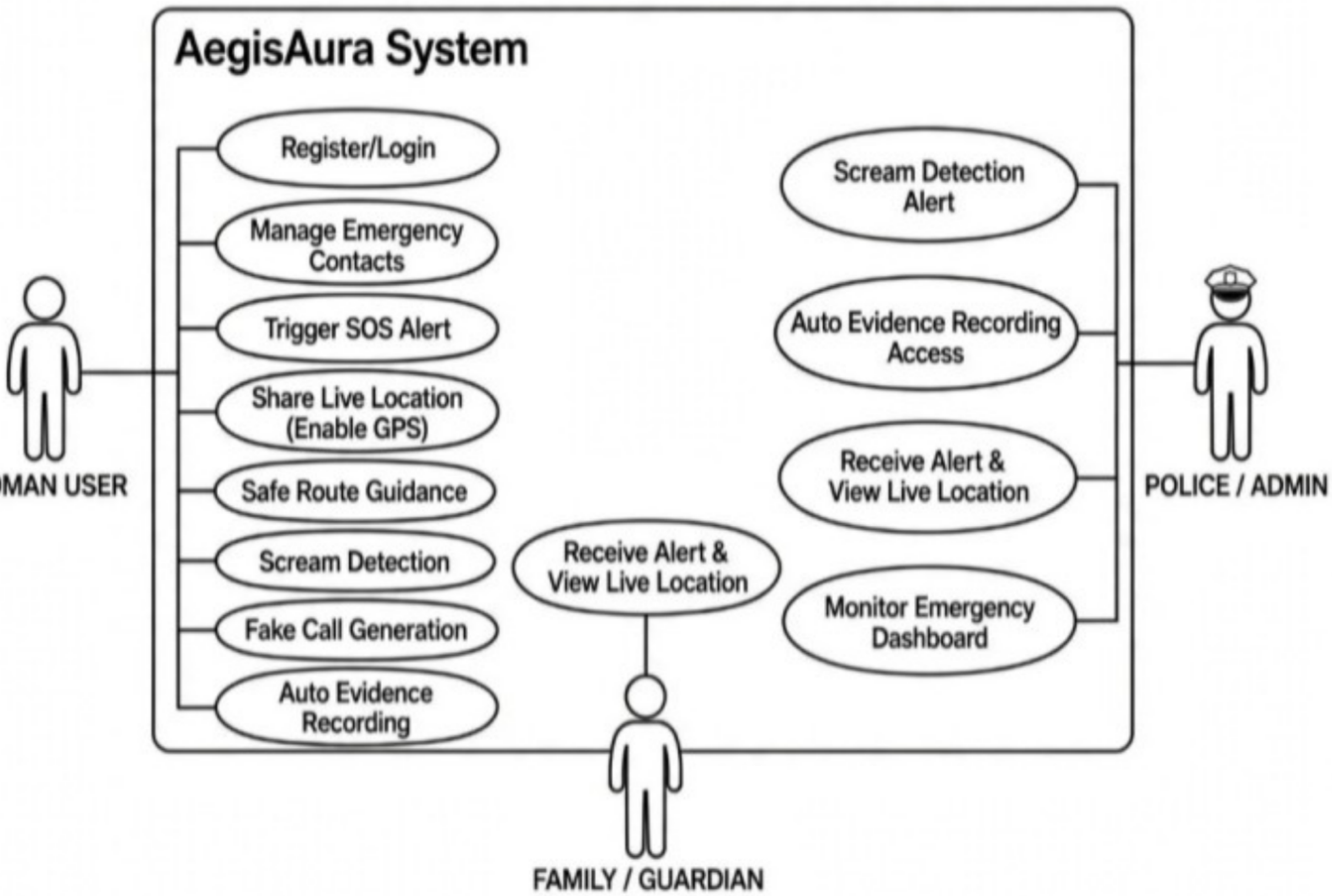
Hands-on activity:

- **User Registration and Login**: The user will open the app and create an account using her mobile number and email. After that she can securely login to the system.
- **Add and Manage Emergency Contacts**: The user will add trusted family members and police helpline numbers as emergency contacts. She can update or delete them anytime.
- **One-Tap SOS Alert**: In an emergency, the user will press the big SOS button. The system will immediately send an SMS and call with live location to all emergency contacts.
- **Shake to Send Alert**: If the user is unable to open the screen, she will shake the phone 2 times. The app will automatically detect the shake and

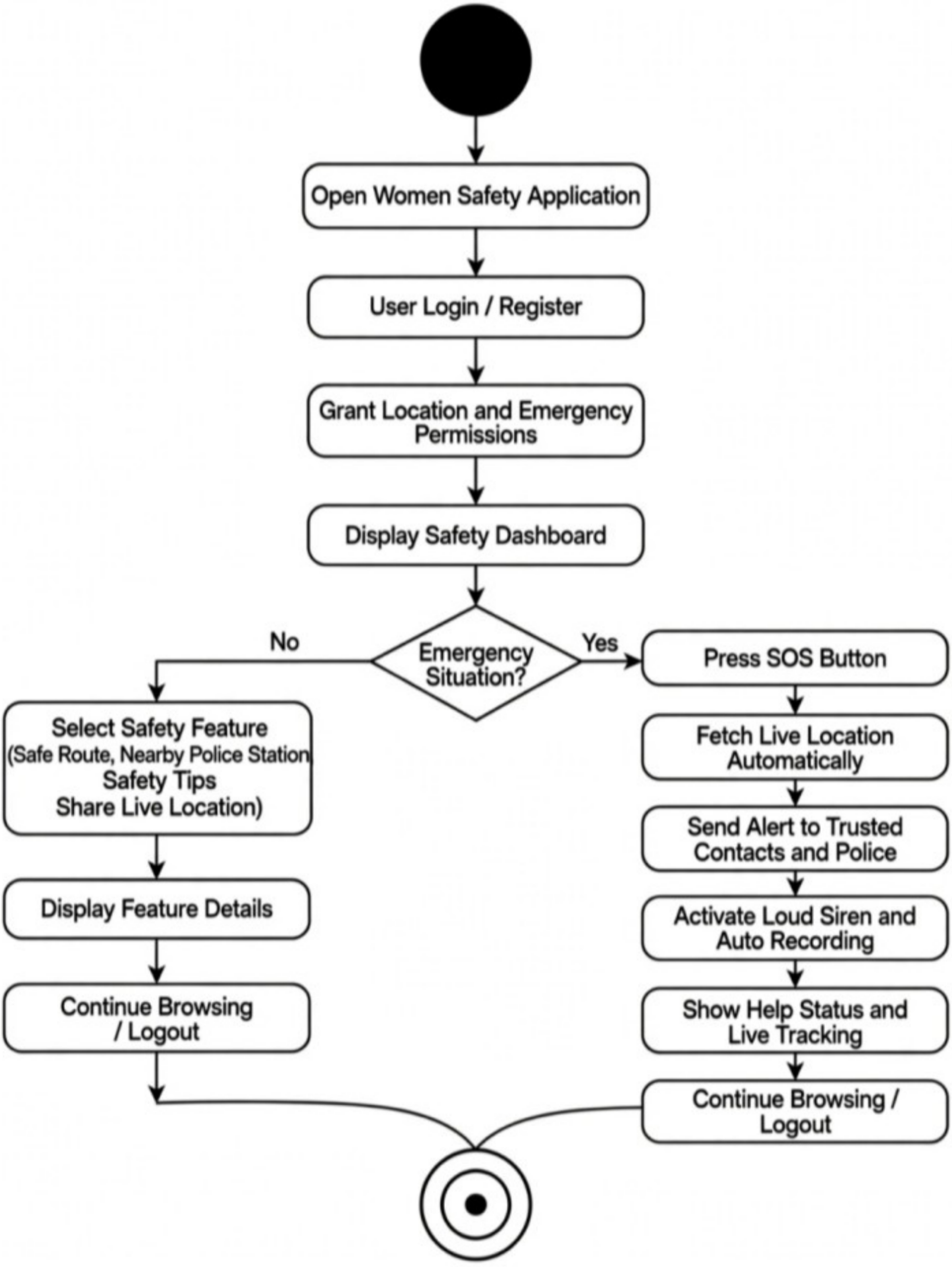
send an SOS without opening the app.

- **Live Location Tracking by Family**: The family members will login with the Guardian account and will see the live movement of the user on the map in real-time.
- **Scream Detection and Auto Alert**: If the user screams loudly, the AI will detect the scream sound and will automatically send an alert to family and police, even if the user does not press any button.
- **Use Safe Route and Fake Call for Safety**: The user will check the map to see which route is safe (green) and which is unsafe (red). If she feels unsafe, she can generate a fake incoming call to escape from the situation.

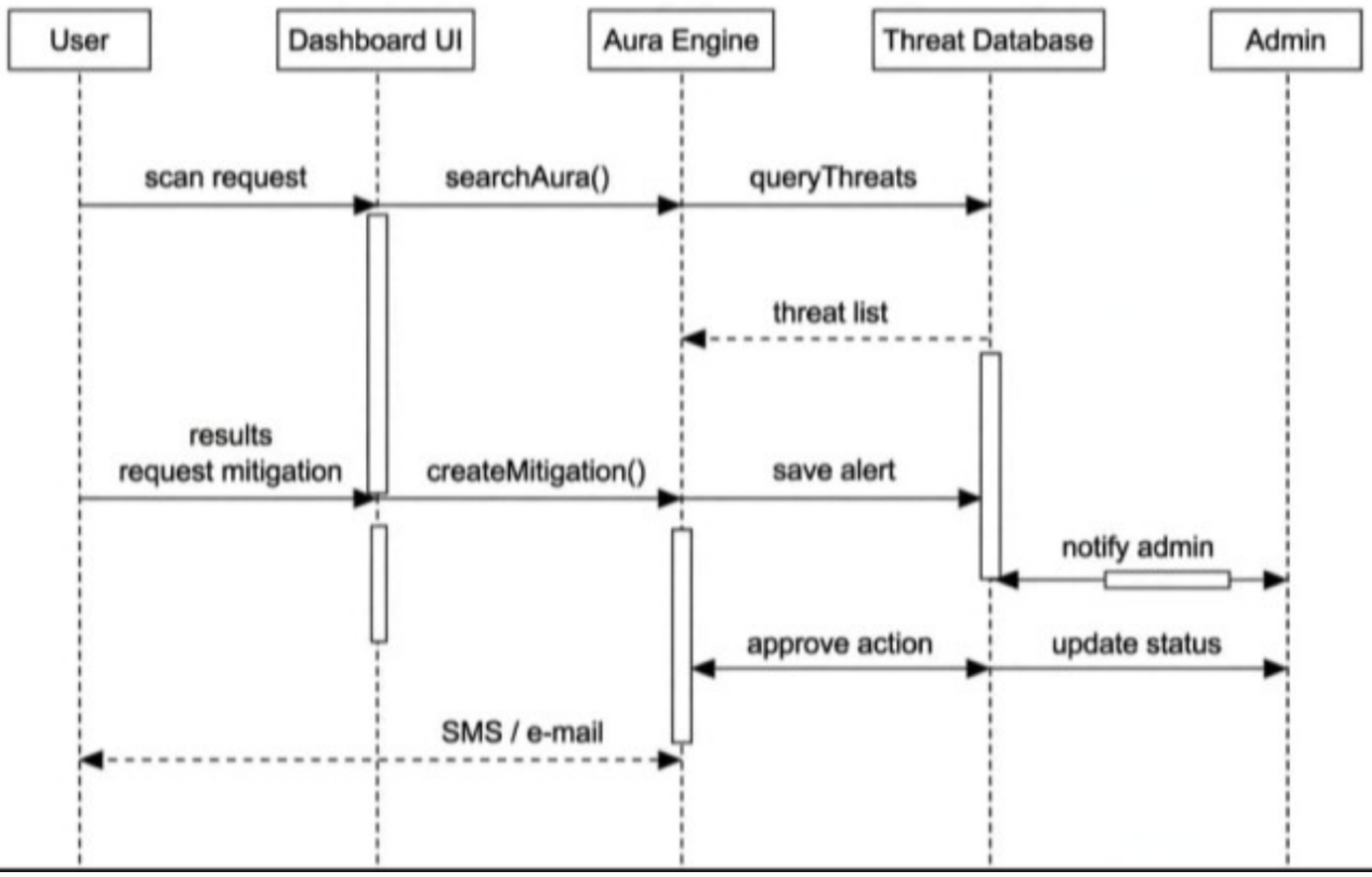
USE CASE DIAGRAM - AegisAura



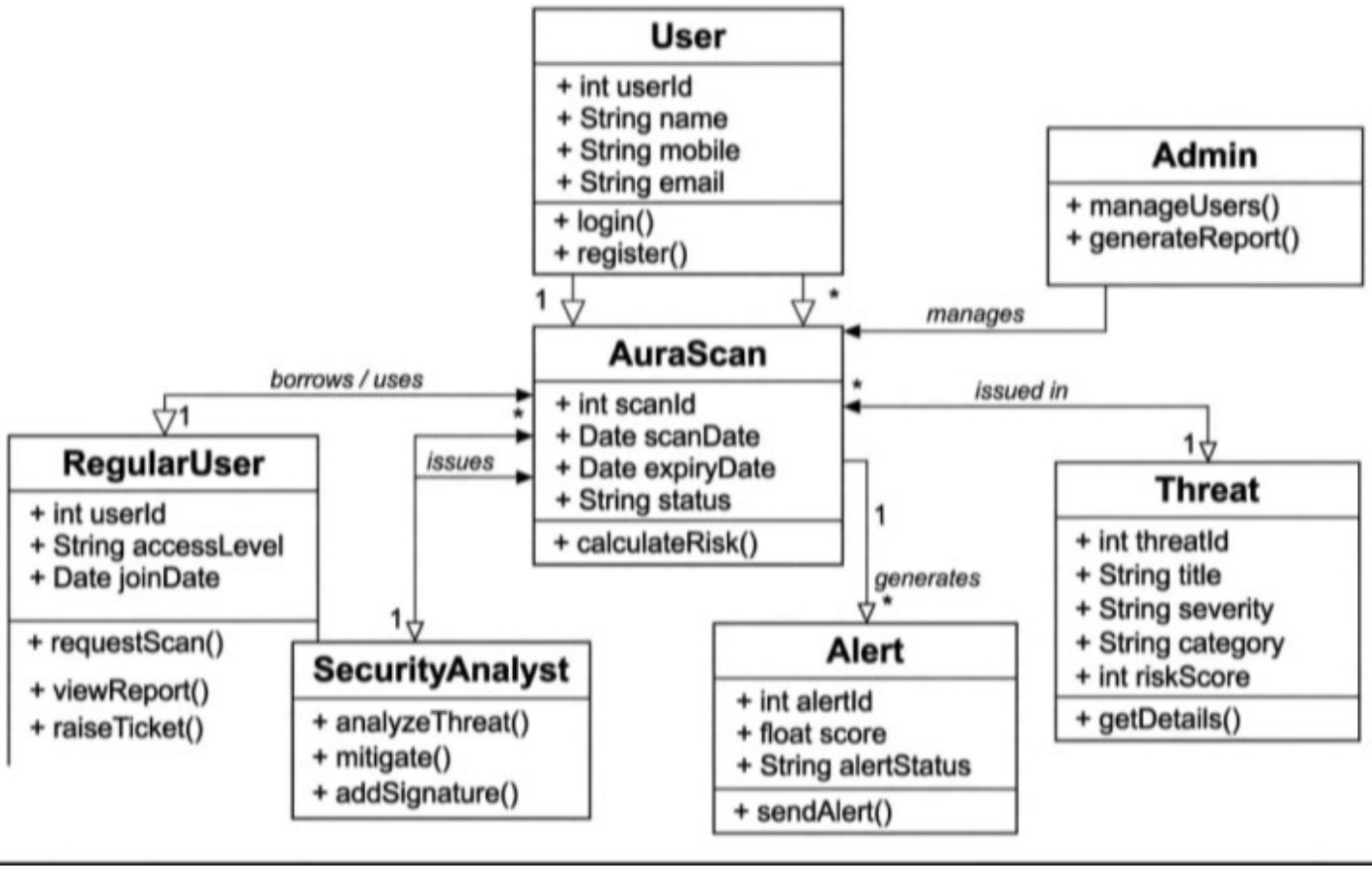
Women Safety App - Activity Diagram



Aegis Aura System - Sequence Diagram



Aegis Aura System - Class Diagram



Collaboration Diagram For Aegis Aura System

